

DHRUV ASHOK PATEL

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MS Cybersecurity Engineering student at USC with hands-on experience in detection engineering, signals development, and cloud threat detection. Built SOC detection pipelines, GCP-based IAM threat detection systems, and LLM agent security research with agentic detection workflows.

Education

Master of Science in Cybersecurity Engineering | University of Southern California Aug 2025 – May 2027
GPA: 3.7/4.0 | Coursework: Computer Forensics (DSCI 528); Malware Analysis & Reverse Engineering (TAC 443); Security & Privacy (DSCI 529); Computer System Assurance (DSCI 523); Security Systems (CSCI 530)

B.E. Computer Science & Engineering (Cybersecurity) | University of Mumbai Dec 2021 – May 2025
GPA: 9.1/10 | Coursework: Digital Forensics, Fundamentals of Information Security, Cybersecurity Risk Management, Information Security Compliance, Ethical Hacking & Cybersecurity, ML for Cybersecurity

Skills

Detection Engineering: Wazuh, Splunk (SPL), Sigma rules, MITRE ATT&CK, Sysmon, custom detection rules, incident response

Cloud & IAM Security: GCP Cloud Audit Logs, GCP IAM, Cloud Security Command Center, Linux, Active Directory

AI/LLM Security: Prompt injection, indirect injection, LLM agent security, agentic detection workflows, MITRE ATLAS

Malware Analysis: Ghidra, x64dbg, scdbg, CyberChef, static/dynamic analysis, shellcode extraction, API hashing

Languages: Python, Bash, PowerShell, C/C++, SQL, JavaScript

Security Tools: Wireshark, Metasploit, Nessus, Nmap, VirusTotal API, Burp Suite

Work Experience

Cybersecurity Intern | Crossover Technology Pvt Ltd. Mar 2025 – Jun 2025

- Reduced alert triage time by monitoring and classifying 50+ daily security alerts across the SIEM dashboard, escalating confirmed threats by severity to the security team for remediation.
- Improved incident documentation accuracy by parsing and analyzing system logs to identify anomalous activity patterns, producing structured reports for the security operations team.

Projects

SOC Detection Engineering Lab | Wazuh, Sysmon, Splunk, Python, MITRE ATT&CK GitHub

- Built detection lab simulating 5 attack scenarios against Windows 11 + Wazuh SIEM; developed 15+ high-fidelity detection signals mapped to 8 MITRE ATT&CK techniques across credential access, defense evasion, persistence, and execution. Automated alert triage pipeline with VirusTotal IOC enrichment; forwarded 500+ alerts into Splunk via HEC with SPL detection queries.

Cloud Threat Detection Lab | GCP Cloud Audit Logs, Splunk, Python, MITRE ATT&CK GitHub

- Simulated IAM privilege escalation on GCP (roles/viewer to roles/owner via SetIamPolicy), detected via Cloud Audit Logs. Developed Python detection pipeline analyzing 32+ audit events with MITRE ATT&CK mapping (T1078.004); configured scheduled automated alerting to reduce manual triage toil.

LLM Agent Security: Indirect Prompt Injection Lab | Python, Anthropic API, React GitHub

- Implemented 5 indirect prompt injection attack classes against an LLM agent on Anthropic's tool-use loop; built 5 defensive mitigations and an eval harness benchmarking all attack/defense combinations. Developed React UI with real-time SSE streaming for live detection visualization, directly applicable to agentic detection engineering workflows.

Quantum-Secured Payment System | Python, FastAPI, React, MongoDB, Docker GitHub

- Built a full-stack payment platform implementing BB84 quantum key distribution with eavesdropper detection, AES-256-GCM transaction encryption, card tokenization, and OTP-verified checkout across card, UPI, and net banking flows.

Malware Detection Using CNN-LSTM | Python, TensorFlow, Flask GitHub | 2024

- Achieved 94.06% accuracy on imbalanced Windows PE dataset using hybrid CNN-LSTM with SMOTE and Keras Tuner; deployed Flask app with confidence threshold slider and saliency map explainability.

Publications

Advancing Security in Digital Transactions using Quantum Cryptography – Journal of Information Systems Engineering and Management (JISEM), Apr 2025

Malware Classification using Deep Learning – International Research Journal of Engineering and Technology (IRJET), Oct 2024

Certifications & Extracurricular

- **Certifications:** Google Cybersecurity Specialization | Cybersecurity for Everyone (UMD)
- **TryHackMe & HackTheBox:** 50+ labs (vulnerability assessment, intrusion detection, digital forensics, privilege escalation)
- **Award:** 1st Place, WordHack Competition (CyberPeace Foundation)